Technical Proposal

D & S Electric Works. Integrated operations platform for engineering-led, project-based manufacturing.

> **Status: NEGOTIATION DRAFT, pending company approval.** This proposal offers a bounded Phase 1 (see the Commercial Proposal). It does **not** offer every module in DSEW's enquiry within that phase.

1. Executive summary

DSEW makes engineered, often one-off products: transformers, magnet coils, custom electromagnets and research equipment. The business risk is not in recording what was made. It is in making sure that what was approved is what was built, that every drawing revision and measurement can be traced afterwards, and that nothing leaves the factory without the required checks.

We propose to start with exactly that core: one job that carries a project from enquiry through quotation, a revision-controlled bill of materials, controlled documents, manufacturing, structured test records and dispatch. Changing an approved record goes through a revision or amendment, never an edit. Quality rules are enforced by the system, not by good intentions.

How we will deliver. We build on Verity, our own multi-tenant business platform, so that tenant isolation, permissions, audit, stock integrity and approvals are not rebuilt for you. We add the DSEW-specific engineering, document and quality capabilities in two-week sprints, each demonstrated on a test system, and deploy on your premises.

What this proposal is not. It is not a promise of all modules A to S of the enquiry for one fixed fee. Phase 1 is a defined core. The remaining modules are future phases, quoted separately once Phase 1 has fixed their depth, and one item (Tally Prime integration) is excluded pending DSEW's written acceptance (section 7).

2. How we describe capability status

None of these labels means "validated against DSEW's workflow". That validation is what Phase 1 does.

Label	Meaning
Foundation present	The core behaviour exists in the platform today and has been demonstrated with fictional data. It has **not** yet been configured or validated for DSEW; that is Phase 1 work where marked included.
Extend	Part of it exists; the DSEW-specific part (revisions, tolerances, your formats) is built in Phase 1 if marked included.
Build	Not present today.
In Phase 1	Included in the fixed Phase 1 price, within the boundary in the Commercial Proposal.
Future phase	Not included in the Phase 1 price; quoted separately.
Excluded, deferred	Not offered. Needs DSEW's written acceptance.

3. Coverage of the enquiry (modules A to S)

Module	Platform status	Phase 1	What Phase 1 delivers, or why not
A. CRM	Extend	Partly	Enquiry register and quotation flow, customer master. Contact directory, RFQ/tender tracker, meetings, email filing are future phase
B. Project management	Foundation present	In Phase 1	Project job, milestones, tasks and dependencies, timesheets, cost lines, quoted versus actual. Risk register and deeper WBS are future phase
C. Enquiry, quotation, sales	Extend	Partly	Quotation revisions, approval, DSEW print format, order and job creation. PO-versus-quote comparison, alternative offers, estimate build-up templates, retention and LD tracking are future phase
D. Engineering and PLM	Extend	Partly	Revision-controlled BOM (single-level) with release and pinned revision per order. Multi-level BOM, full ECR/ECO and design reviews are future phase
E. Document management	Build	Partly	Controlled documents with one numbering convention, revisions, states, register, permissions, audit. Transmittals, comparison, retention rules are future phase
F. Manufacturing execution	Foundation present	Partly	Orders from a released BOM, routes, operations, hold and resume, material issue. Machine allocation, operator time, subcontracting, planned-versus-actual consumption are future phase
G. Travellers	Extend	Partly	Printable traveller in a DSEW format, QR identity of a finished unit. Hold and witness points beyond pass/fail checks, customer-facing variants, lot identification are future phase
H. Quality management	Extend	Partly	Quality gates, failed-or-missing check blocking, authorised audited override. QAP/ITP, NCR/CAPA, audits, FAI/FAT packs are future phase
I. Barcode, QR, traceability	Extend	No	Future phase (the QR identity of a finished unit is included under G)
J. Procurement, vendors	Extend	No	Future phase
K. Inventory, warehouse	Extend	Foundation only	Existing stock behaviour is used as is. Extensions are future phase
L. Production planning, MRP	Build	No	Future phase
M. Testing, equipment	Build	Partly	Test templates for up to 3 product families, readings with unit and tolerance, automatic pass/fail, report, basic instrument register with calibration-due check. Maintenance schedules, FAT packs, customer certificates beyond one format are future phase
N. Finance and Tally Prime	n/a	**Excluded, deferred**	**No Tally Prime integration. Requires DSEW's written acceptance (section 7).** Project cost and quoted-versus-actual visibility is included under B
O. HR and payroll	Extend	No	Future phase

Module	Platform status	Phase 1	What Phase 1 delivers, or why not
P. Dispatch, logistics	Extend	Partly	Dispatch record, packing list, delivery challan, partial shipment, gated on completion and tests. E-way bill, warranty tracking are future phase
Q. Contracts, compliance, risk	Build	No	Future phase
R. Dashboards and BI	Extend	Partly	Four read-only views (quotations, jobs, production, quality). Role dashboards and exports are future phase
S. Notifications, workflow	Extend	Partly	Existing approval chains. Escalations, e-mail, WhatsApp, rule configuration are future phase
AI	n/a	No	Future phase (section 5)
Hindi	n/a	No	Future phase

4. Engineering, documents and quality: the core of Phase 1

These three areas decide whether the system is trusted on the shop floor and by your customers' inspectors.

Revision-controlled BOMs (D). A BOM is never overwritten. A new revision is drafted, reviewed with a stated reason, and released. A released revision is immutable. A manufacturing order records exactly which revision it was built to, so a change next month cannot rewrite a job from last month. The system refuses to start production on a BOM that is not released.

Controlled documents (E). Each document has a number from your convention, a revision, a state and an owner. The register shows what is current and what is superseded. Releasing is a recorded event. Controlled engineering documents are kept distinct from general photos and attachments, so evidence cannot be mistaken for an approved drawing.

Structured testing (M). A test record holds real readings: value, unit and tolerance, with tester and reviewer. Pass or fail is computed from the acceptance limits on the server, so it cannot be bypassed by another screen or a direct call. A missing or failed required check blocks completion and dispatch. An authorised override needs a stated reason and is permanently audited.

5. Artificial intelligence (future phase)

Not part of Phase 1. For planning only: if DSEW later wants AI, it would be an assistant, never an approver, able to run on local models on DSEW's own server within each user's permissions. It would not approve, release or authorise anything, and no essential workflow would depend on it. Scope and price would be set only if DSEW asks.

6. User interface

- A modern, responsive interface for office PCs and for phones and tablets on the shop floor.
- Searchable, sortable, filterable lists, and links between related records: a project opens to its customer, quotation, order, BOM, job, traveller, tests and documents.

- Changes to approved records follow the revision or amendment path; the interface offers the correct path rather than a blocked edit.
- Printable, branded documents in Phase 1: quotation, traveller, test report, packing list, delivery challan.
- English in Phase 1. Hindi is a future phase.
- Current Chrome, Edge and Firefox on Windows. Accessible contrast and 44-point touch targets on phones.

7. Requirement of the enquiry not covered: Tally Prime

DSEW's enquiry (section N) asks for Tally Prime integration and for the integration method, licensing, supported version, synchronisation frequency and limitations to be specified.

This proposal excludes Tally Prime integration and defers it. It does not specify a method, version or frequency, does not price it, and does not claim to meet that requirement. Tally Prime remains DSEW's accounting system and book of record; Verity neither reads nor writes Tally in Phase 1. DSEW must confirm in writing that it accepts this exclusion. If DSEW requires integration, it would be scoped and quoted separately after a technical review of DSEW's TallyPrime setup, and only on explicit authorisation.

8. Technical architecture

Layer	Choice	Licence cost
Application	TypeScript, Next.js and React web application	Open source, none
Data	PostgreSQL 16, row-level security isolating each company's data	Open source, none
Identity	Keycloak (OpenID Connect). Multi-factor sign-in is a Keycloak feature; enabling and testing it for DSEW users is part of set-up if DSEW asks for it, and it is not enforced by default	Open source, none
Files	S3-compatible object storage (SeaweedFS), checksum on every stored file	Open source, none
Packaging	Docker containers, install, upgrade and backup scripts	Open source, none
Reverse proxy and TLS	Caddy or nginx with your internal certificate authority	Open source, none

No per-user, per-module or subscription licence is required to run the platform. A licence inventory of every third-party component is part of the handover.

Data export. Phase 1 includes one documented export of DSEW's data: tabular records as CSV files and uploaded files in original form with an index, taken at handover and once more on request during warranty, with record and file counts reconciled as the acceptance test. The exact fields and files are listed

in the specification.

Integration and API. A permissioned REST interface with API keys tied to a user, so an integration can never exceed that user's permissions. No external integrations are included in Phase 1.

9. Deployment on your infrastructure

Windows Server 2019: our direct answer

Verity's runtime is Linux-based. We **do not** propose running it natively on Windows Server 2019, and we do not recommend Docker Desktop or WSL for production there: WSL 2 is not a supported production path on Server 2019, and Windows containers cannot run our Linux components.

Proposed approach. Run one Linux virtual machine (Ubuntu Server LTS) on your Windows Server 2019 host using the Hyper-V role. The VM holds the application, PostgreSQL, file storage and Keycloak. The Windows host is not modified beyond enabling Hyper-V. If the host is too small or busy, the alternative is a small dedicated server running the same VM. Licensing of the Windows host is for your Microsoft partner to confirm; the Linux guest carries no licence cost.

What we have verified and what we have not. We have run the install, schema migration, identity sign-in and file upload paths on our own Linux test environments. We have **not** yet completed a combined rehearsal of database, files and identity restore on a clean second environment, tested a real phone uploading over HTTPS through the proxy, or run on your Windows Server 2019. Those are explicit acceptance steps, to be passed on a test VM before go-live. We will not commit a date for installation on your premises until your server facts are surveyed.

Your 15 questions

#	Question	Answer
1	Architecture and stack	Section 8
2	Direct on Windows Server 2019?	No. Linux VM under Hyper-V, or a small separate host
3	Recommended hardware	For Phase 1 (15 named users): 4 vCPU, 16 GB RAM, 500 GB mirrored SSD, a separate backup target at least equal to data size, UPS, gigabit LAN. Confirmed after the survey
4	Database and licence	PostgreSQL 16, open source, no licence fee
5	Browsers and devices	Current Chrome, Edge, Firefox on Windows; Android and iOS phones and tablets
6	Network and remote access	LAN inside the plant. Remote access through your VPN. We do not expose the server to the internet by default
7	Backup and recovery	Backup scripts for the database and files exist; an identity-configuration backup is part of the procedure. **The combined restore onto a clean environment has not yet been rehearsed and is a go-live gate** (Commercial Proposal, section 3). Recovery time and data-loss window to be agreed with you

#	Question	Answer
8	Security, authentication, audit	TLS, role-based permissions, per-company data isolation enforced in the database (tested by the platform's isolation tests), API keys bound to a user, MFA as described in section 8. The audit trail is append-only for the application: the application's database role cannot change or delete audit records, as verified by tests. A database administrator with elevated rights is outside that guarantee
9	Encryption and remote access	TLS in transit; backup encryption and disk encryption are recommended set-up steps, confirmed at acceptance; VPN for remote use
10	Performance and growth	Design target for Phase 1 is 15 named users; no load test has been run for DSEW yet, and one against an agreed workload is part of acceptance
11	Offline operation	Normal in-LAN operation after deployment needs no internet. Installation, upgrades and any support access need either controlled internet access or offline media. Browser-offline data entry on phones is not in Phase 1
12	Integration and API	Section 8. No external integrations in Phase 1; Tally Prime is excluded (section 7)
13	Repository and version control	A private repository for DSEW-specific work, shared from the start; platform source terms are unresolved (document 4)
14	Test, staging, production	A company-hosted environment for sprint demonstrations, and **one** DSEW-hosted Linux VM used for UAT and then, after the go-live gates pass, as production. A separate staging or second production VM is not included and would be a change request
15	Upgrades and long-term support	Versioned releases, tested upgrade and rollback, migration safety checks. Support after warranty is under an AMC to be quoted

10. Security summary

- **Tenant isolation in the database**, not only in application code, tested on every change.
- **Authorisation on the server for every action.** Hiding a button is never the only control.
- **Audit trail** records actor, time, record, field and reason. The application's database role cannot change or delete it (verified by tests); a database administrator with elevated rights is outside that guarantee.
- **Reversible access suspension** without deleting history, effective at the next request. Implemented and tested on company test databases; accepted for DSEW during UAT.
- **Secrets** are never stored in the repository.
- **Controlled egress.** The design intends no outbound connection unless a feature you enabled needs one. This is **not yet verified** for DSEW; an outbound-network restriction test is part of acceptance.
- **Graceful degradation.** The design keeps existing sessions working if the identity service is briefly down, fails new sign-ins and uploads with a clear message if a service is down, and never relaxes quality gates. The identity and object-store outage tests have **not yet been run** and are part of acceptance.